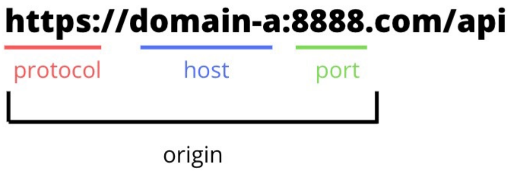
CORS error 

- CORS (**Cross-Origin Resource Sharing**) is a **browser security mechanism**.
- It controls whether a **webpage** can **request resources from** a **different origin.**
- The **browser blocks** the **response** if **CORS headers** are **missing**.
- An **origin** = **protocol + domain + port**.
- Example
  http://localhost:3000 → https://api.com = cross-origin



Why CORS is Needed ?

- **Browsers** enforce **Same-Origin Policy**.
- **Prevents** malicious websites from accessing sensitive data using your logged-in session.
- **Servers** must **explicitly** allow **other origins** to access their resources.

```
Access to fetch at 'http://localhost:5000/gfg-articles' from origin 'http://127.0.0.1:5500' has been blocked by CORS policy: No 'Access-Control-Allow-Origin' header is present on the requested resource. If an opaque response serves your needs, set the request's mode to 'no-cors' to fetch the resource with CORS disabled.   index.html:1
```

How CORS Works ?

- **Browser** sends **request** with an **Origin header**
- **Server** must **respond** with **CORS headers**
- **Browser checks** those **headers**
- If **allowed** → **response** accessible
- If **not** → **CORS error**

```
Access-Control-Allow-Origin: http://localhost:3000
```

OR

```
Access-Control-Allow-Origin: *
```